

1. Administrative & Study Identification

Full Study Title An Efficacy, Safety, Tolerability, Immunogenicity, and Lot-Consistency Clinical Trial of a 6-Valent OspA-Based Lyme Disease Vaccine (VLA15) **Short Title/Acronym** VALOR (Vaccine Against Lyme for Outdoor Recreationists) **Protocol Number** C4601003 **NCT Number** NCT05477524 **Sponsor Name** Pfizer Inc. / Valneva SE **Investigational Product** PF-07307405 (LB6V, formerly known as VLA15) **Phase of Development** Phase III **Medical Monitor** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of a 6-valent OspA-based Lyme disease vaccine (VLA15) for the prevention of confirmed Lyme disease compared to placebo. **Secondary Objectives:** To evaluate the safety, tolerability, and immunogenicity of VLA15; to establish lot-consistency across 3 different lots; and to determine immunobridging non-inferiority between pediatric (5-17 years old) and adult (18-44 years old) participant strata.

2.2 Background & Rationale Lyme disease is the most common vector-borne illness in the Northern Hemisphere, with an estimated 476,000 Americans and 132,000 Europeans diagnosed and treated annually. It is caused by *Borrelia* bacteria transmitted through the bite of infected *Ixodes* (blacklegged) ticks. Currently, there is no approved human vaccine available on the market. This study aims to evaluate VLA15, which specifically targets the outer surface protein A (OspA) of six *Borrelia* serotypes, neutralizing the pathogen within the tick to interrupt transmission to the human host.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Saline) **Blinding:** Observer-blinded / Double-blind **Randomization:** 1:1 ratio (VLA15 or placebo); a subset is randomized in a 1:1:1:3 ratio to assess lot equivalence.

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 9,400 healthy participants **Number of Sites:** Multicenter across North America and Europe in areas with high levels of endemic Lyme disease **Estimated Study Start:** August 2022 **Estimated Primary Completion:** December 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy male and female participants, age ≥ 5 years and older. 2. Reside in or frequently visit areas with high levels of endemic Lyme disease (e.g., individuals who work in forestry, landscaping, or pursue outdoor recreational activities). 3. High risk of exposure to *B. burgdorferi*-infected ticks.

4.2 Exclusion Criteria 1. Newly developed or unstable underlying conditions that may interfere with the assessment of Lyme disease (e.g., chronic arthralgia/arthritis, chronic skin conditions). 2. Any unstable autoimmune condition with manifestations (e.g., arthritic and neurologic) that may interfere with the assessment of Lyme disease. 3. Chronic systemic doxycycline, minocycline, or other tetracycline class drug use for acne or other conditions.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 3-dose primary vaccination series administered at Months 0, 2, and 5 to 9, followed by a 4th booster dose approximately 12 months after the primary series (ahead of the second Lyme disease season). **Route of Administration:** Intramuscular injection. **Duration of Treatment:** 4 total doses over ~1.5 years; observation and follow-up through study completion up to approximately 42 months (spanning a third Lyme disease season).

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowable concomitant medications. **Prohibited:** Chronic systemic suppressive antibiotics (e.g., tetracycline class drugs) that could mask or interfere with Lyme disease infection and symptom presentation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening/Baseline	Month 0	Informed Consent, Medical History, Randomization, Dose 1
Interim 1	Month 2	Dose 2 Administration, Safety & Tolerability Assessment
Interim 2	Months 5 to 9	Dose 3 Administration, Safety, Immunogenicity (Blood Draw)
Interim 3	Months 17 to 21	Booster (Dose 4) Administration ahead of 2nd Lyme season
End of Study	Up to 42 Months	Final Efficacy Assessment, Exit Interview, Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Rate of confirmed Lyme disease cases from 28 days post-dose 4 (and 1 day post-dose 4) compared

to the placebo arm. **Measurement Method:** Clinical diagnosis by the investigator (e.g., erythema migrans rash) combined with laboratory confirmation (serology/PCR for *Borrelia burgdorferi*).

7.2 Secondary & Exploratory Endpoints 1. Safety and tolerability profile (occurrence of local/systemic adverse events). 2. Immunogenicity: *Borrelia* OspA-specific antibody titers and seroconversion rates. 3. Lot-consistency across three independent manufacturing lots.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient/guardian e-diaries utilized for tracking local and systemic reactions post-vaccination, alongside routine clinical surveillance. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) utilized to oversee safety and trial integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population, Per-Protocol (PP) set for primary efficacy, and Safety Set. **Sample Size Justification:** Approximately $N=9,400$ adequately powered to demonstrate vaccine efficacy. Pre-determined statistical success criterion required the lower bound of the 95% confidence interval for efficacy to exceed 20%. **Handling of Missing Data:** Standard multiple imputation and time-to-event survival analysis censoring methodology for unconfirmed cases.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote monitoring (Note: 8 sites were terminated during the trial for GCP/quality issues to preserve trial integrity). **Audit Trail:** eCRF locking and adjudication of clinical endpoints by an independent review committee.

11. Study Identifiers & Governance

Primary ID: {{C4601003}} **Registry Number:** {{NCT05477524}} **Sponsor/Lead Organization:** {{Pfizer Inc. / Valneva SE}} **Study Title:** {{VALOR}} **Official Regulatory Title:** {{An Efficacy, Safety, Tolerability, Immunogenicity, and Lot-Consistency

12. Clinical Framework (Lyme Disease Specific)

Primary Condition: {{Lyme Disease}} **Diagnosis Threshold:** {{Clinical manifestations (e.g., erythema migrans) + laboratory confirmation of Borrelia burgdorferi}} **Medical Methodology:** {{Prophylactic Vaccination (6-Valent OspA-Based Vaccine)}} **Optimization Target:** {{Prevention of Borrelia transmission from tick to human}}

13. Study Procedures & Methodology

Management Pathway: {{Vaccine Administration and Active Disease Surveillance}} **Education Protocol (HCP):** {{Tick-borne disease prevention, early symptom recognition, and vaccine administration protocol}} **Education Protocol (Patient):** {{Tick bite prevention, recognition of erythema migrans rash, and eDiary reporting}} **Quality Audit Mechanism:** {{Safety and Immunogenicity monitoring, Clinical Endpoint Adjudication}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 42 months}} **Onsite Visit Cadence:** {{Months 0, 2, 5-9, and Booster at ~17-21 months}} **Remote Monitoring Frequency:** {{Active surveillance during peak Lyme disease seasons}} **Checklist Review Interval:** {{Routine eDiary completion for symptoms and tick bites post-vaccination}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				